

# Boundless Skies

## # Executive Summary

Boundless Skies is an accessible astronomy charity that operates the world's first automated telescope network designed from the ground up for people with disabilities. Seestar smart telescope owners donate their telescope's nights to the network. Our AI automatically schedules observations of scientifically valuable targets, processes the data into calibrated photometry, and submits it to professional astronomical databases. Members contribute to real science — supernovae, novae, variable stars, transient phenomena — without physical effort, technical expertise, or significant cost. The entire experience is accessible via a mobile app built specifically for users with visual, motor, and cognitive disabilities.

The night sky belongs to everyone. Boundless Skies makes that true.

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## # The Problem

Astronomy has an accessibility crisis that nobody talks about loudly enough.

Traditional amateur astronomy requires you to carry heavy equipment to dark locations, stand in the cold for hours, operate fine controls with precision, look through an eyepiece with functional vision, and travel to sites inaccessible to wheelchairs. Every one of these requirements excludes someone.

Existing citizen science programs offer classifications of existing images — clicking on galaxies, identifying craters. This is not astronomy. This is data labeling. Contributors are not named on scientific outputs. They do not collect original data. They are not astronomers.

Professional telescope networks like Las Cumbres Observatory are inaccessible to the public. Consumer smart telescope networks like Unistellar are closed platforms with proprietary hardware costing thousands of dollars and no accessibility focus.

The result: disabled people are almost entirely excluded from active astronomical participation despite astronomy being the science most naturally suited to remote, instrument-mediated participation.

Nobody has solved this. Not because they tried and failed. Because nobody tried.

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## # The Solution

Boundless Skies is a three-part system:

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1. A distributed network of donated Seestar telescopes operating autonomously overnight
2. An AI-driven pipeline that decides what to observe, processes the data, and submits it to science
3. A fully accessible mobile app that gives anyone — regardless of physical ability — a genuine connection to the science their telescope produces

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Members contribute their telescope's nights to the network. The network does the science. The science is real, credited, and permanent.

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## # How It Works

### ## The Member Experience

#### ### Joining

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1. Download the Boundless Skies app (free)
2. Create an account (free)
3. Enable Station Mode on your Seestar  
(one-time, 5-minute setup)
4. Download and run the Boundless Skies  
Node Software on your computer  
(one-click installer)
5. Your node is live

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Total setup time: approximately 15 minutes. No technical knowledge required. No hardware purchase. No monthly fee.

### ### Every Night

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The member does nothing.

At astronomical twilight:

- Node software wakes up automatically
- Downloads tonight's observation plan from Boundless Skies cloud
- Opens connection to Seestar

During the night:

- Seestar slews to each target
- Takes stacked exposures
- Images transfer to member's computer via local network
- Photometry runs automatically
- Measurements upload to Boundless Skies
- Repeat

At dawn:

- Seestar parks
- Night summary generated
- Member notified of contributions

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### ### Every Morning

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Member receives notification:

"Last night your telescope observed 6 targets across 4 hours.

Your data confirmed SN2025abc is consistent with a Type Ia supernova.

Your name has been added to the observation record in the AAVSO database.

View your results →"

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## ## The Technical Pipeline

### ### Layer 1: Alert Ingestion

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#### SOURCES:

- ALeRCE (ZTF alerts via Vera Rubin brokers)
- ATLAS forced photometry server
- ASAS-SN bright transient server
- AAVSO alert notices
- Gaia Science Alerts
- TNS (Transient Name Server)

#### PROCESS:

- Continuous polling of alert APIs
- Parse incoming alerts
- Deduplicate across sources
- Cross-match against known source catalogs
- Initial brightness filter (mag < 14)
- Store in target database

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### ### Layer 2: Scoring and Prioritization

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#### SCORING FUNCTION:

Score = brightness\_match  
    × scientific\_value  
    × time\_criticality  
    × coverage\_gap  
    × observability

#### BRIGHTNESS MATCH:

Peak score for mag 10-13  
(optimal for Seestar precision)  
Penalty below mag 8 (saturation risk)  
Penalty above mag 14 (detection limit)

#### SCIENTIFIC VALUE:

Classification probability from brokers  
Target type weighting:  
Classical nova: 1.0 (highest)  
Type Ia supernova: 0.95

CV outburst: 0.9  
Core-collapse supernova: 0.85  
AGN flare: 0.7  
Mira at maximum: 0.6  
Known eclipsing binary: 0.5

#### TIME CRITICALITY:

How fast does value decay?

Kilonova: hours

Nova: days

Supernova: days to weeks

CV outburst: days

Mira: weeks

Eclipsing binary: fixed schedule

#### COVERAGE GAP:

Is anyone else observing this?

If professional facilities are on it,  
deprioritize unless we add unique value  
(continuous coverage, longitude gap)

#### OBSERVABILITY:

Altitude > 30° from each node

Moon distance > 30°

Weather probability at node location

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#### ### Layer 3: Scheduling

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#### INPUTS:

- Ranked target list
- Node registry (location, capability, status)
- Weather forecasts per node location
- Current time and ephemeris data

#### ALGORITHM:

Priority queue with greedy dispatch

For each node, for each night:

1. Calculate visibility window per target
2. Estimate exposure time needed per target
3. Assign highest-scoring visible targets
4. Fill remaining time with steady-state program

5. Generate ordered observation plan
6. Send to node via API

#### INTERRUPT HANDLING:

High-priority alerts can interrupt

current observations:

- Nova discovery: interrupt after current exposure
- Kilonova candidate: immediate interrupt
- Normal targets: complete current observation

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#### ### Layer 4: Node Software

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RUNS ON: Member's existing Windows/Mac/Linux computer

INSTALLS AS: Background service (invisible to member)

COMMUNICATES WITH: Seestar via local WiFi

Boundless Skies cloud via internet

#### MAIN LOOP:

1. Connect to Seestar via seestar\_alp (Alpaca API)
2. Download observation plan from cloud
3. For each target:
  - a. Send slew command to seestar\_alp
  - b. Wait for slew and plate solve (~30-60 sec)
  - c. Send exposure command
  - d. Wait for completion (~2-5 min)
  - e. Detect new image on SMB share
  - f. Run local photometry pipeline
  - g. Upload measurement to cloud (~1 KB)
  - h. Report status to cloud
4. At dawn: park Seestar
5. Send night summary to cloud

#### SAFETY:

- Heartbeat to cloud every 60 seconds
- Auto-park if cloud connection lost > 30 min
- Auto-park at astronomical dawn regardless
- Graceful handling of Seestar disconnects
- Local log of all actions for debugging

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#### ### Layer 5: Local Photometry Pipeline

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RUNS ON: Member's computer (background)

TOOLS: Astropy, Photutils, ASTAP

FOR EACH IMAGE:

1. PLATE SOLVE

- ASTAP local plate solver
- Produces WCS solution (sky coordinates for every pixel)
- Typical time: 5-15 seconds

2. IDENTIFY TARGET

- Match alert RA/Dec to image WCS
- Confirm target is in field
- Flag if target is missing

3. FIND COMPARISON STARS

- Query AAVSO comparison star sequence (for known targets)
- Or query Gaia DR3 / ATLAS Refcat2 (for new transients)
- Select 5-10 comparison stars: similar brightness, similar color, not variable, in same field

4. APERTURE PHOTOMETRY

- Measure flux of target + comparison stars
- Aperture radius: 2-3× FWHM
- Background estimated from annulus
- Instrumental magnitude for each

5. DIFFERENTIAL PHOTOMETRY

- Calculate zero point from comparison stars
- Apply to target
- Propagate uncertainties

6. QUALITY METRICS

- SNR of target
- FWHM (seeing estimate)
- Airmass (from coordinates + time)
- Background level
- Number of comparison stars used
- Quality flags

## 7. OUTPUT

```
{
  target_name: "SN2025abc",
  bjd: 2460500.1234,
  magnitude: 13.42,
  uncertainty: 0.08,
  filter: "CV",
  airmass: 1.34,
  fwhm: 3.2,
  snr: 45,
  comparison_stars: 7,
  quality_flag: "good",
  node_id: "node_042"
}
...
```

### ### Layer 6: Cloud Processing and Validation

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RECEIVES: Photometry measurements from all nodes

#### CROSS-VALIDATION:

If multiple nodes observed same target same night:

- Do measurements agree within uncertainty?
- Flag outliers for review
- Weight by SNR for combined measurement
- Flag systematic offsets between nodes

#### LIGHT CURVE MANAGEMENT:

- Store all measurements in database
- Generate light curves per target
- Detect anomalies (unexpected brightening, rapid fading, periodic behavior)
- Alert professional astronomers if warranted

#### QUALITY CONTROL:

- Automated acceptance criteria:
  - SNR > 20
  - Uncertainty < 0.3 mag
  - At least 3 comparison stars
  - Airmass < 3.0
  - Quality flag: good or acceptable
- Failed criteria: flagged, not submitted



→ Borderline: human review queue

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### ### Layer 7: Data Submission

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#### AAVSO SUBMISSION:

Format: AAVSO Extended File Format

#TYPE=EXTENDED

#OBSCODE=[network observer code]

#SOFTWARE=Boundless Skies Pipeline v[X]

#DELIM=,

#DATE=BJD

#OBSTYPE=CCD

Fields: STARID, DATE, MAGNITUDE, MAGERR,  
FILTER, TRANSFORMED, MTYPE, CNAME,  
CMAG, KNAME, KMAG, AMASS, GROUP,  
CHART, NOTES

Submission: Via AAVSO WebObs API

Frequency: Daily batch submission

Review: Automated with human spot-check

#### TNS REPORTING:

For significant transient detections:

→ Format per TNS reporting standards

→ Submitted when network data  
adds novel information

#### INTERNAL ARCHIVE:

→ All raw measurements stored permanently

→ FITS images stored for 30 days

(then deleted to save storage)

→ Photometry results stored permanently

→ Publicly queryable via Boundless Skies API

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### # The Science Program

### ## Primary Program: Bright Transient Follow-up

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#### TARGETS:

- Classical novae (mag 4-12 at peak)
- Supernovae in nearby galaxies (mag 10-16)
- Cataclysmic variable outbursts (mag 10-15)
- AGN flares (mag 12-15)

#### CADENCE:

- High priority: every clear night, multiple observations per night
- Standard priority: every clear night
- Low priority: weekly or opportunistic

#### VALUE:

- Dense light curve sampling
- 24-hour continuous coverage via geographic node distribution
- Rapid response (<1 hour from alert)
- Data complementary to professional facilities (different time windows, continuous monitoring they can't provide)

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### ## Secondary Program: Variable Star Monitoring

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#### TARGETS:

- Long-period variables (Miras, semi-regulars)
- Eclipsing binaries (timing program)
- Cataclysmic variables (quiescent monitoring)
- Be stars (outburst monitoring)
- Active galactic nuclei (long-term baseline)

#### VALUE:

- Long baseline data unavailable elsewhere
- Dense cadence fills gaps in AAVSO coverage
- Eclipse timing builds O-C diagrams
- Quiescent CV monitoring catches outbursts early

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### ## Steady-State Filling Program

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WHEN: No high-priority targets available  
TARGETS: Curated list of AAVSO targets  
          needing regular monitoring  
PURPOSE: Keep telescopes productive  
          every clear night  
VALUE: Consistent long-term dataset  
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# The Accessible App

## Design Philosophy

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PRINCIPLE 1: DISABILITY-FIRST DESIGN

Not: "how do we add accessibility features?"

But: "how do disabled people experience  
      astronomy and what do they need?"

PRINCIPLE 2: MULTIPLE MODALITIES

Every piece of information available as:

- Visual (for sighted users)
- Audio (for blind users)
- Haptic (for deaf-blind users)
- Text (for all users, screen readers)

PRINCIPLE 3: NO REQUIRED PRECISION

Every interaction achievable with:

- One finger, one tap
- Voice command
- Single switch
- Eye gaze

PRINCIPLE 4: PATIENCE

No time limits. Ever.

No auto-advancing content.

User controls the pace entirely.

PRINCIPLE 5: PLAIN LANGUAGE

Astronomy explained, not assumed.

Jargon defined when used.

Reading level adjustable.

Audio available for all text.

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## ## Core Screens

### ### Home Screen

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## YOUR TELESCOPE

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### ● ONLINE — Observing

Currently watching:

Nova in Hercules

Brightness: 8.3 magnitude

Time remaining: 23 minutes

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### LAST NIGHT

6 observations completed

[View results →](#)

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### NETWORK

52 telescopes observing now

Across 18 countries

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[ HEAR THIS PAGE]

...

### ### Results Screen

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## YOUR OBSERVATIONS

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Nova Herculis 2025

Jan 15 — Jan 20 (5 nights)

[CHART: Light curve showing  
brightness over 5 days]

[VIEW AS TABLE]

[HEAR DESCRIPTION]  
[FEEL AS HAPTIC PATTERN]

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Jan 20:  $8.3 \pm 0.06$  mag  
Jan 19:  $7.9 \pm 0.05$  mag  
Jan 18:  $7.4 \pm 0.04$  mag  
Jan 17:  $6.8 \pm 0.04$  mag  
Jan 15:  $5.2 \pm 0.03$  mag  
(fading)

Submitted to AAVSO ✓

### Impact Screen

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YOUR CONTRIBUTION

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TOTAL OBSERVATIONS: 847

TARGETS OBSERVED: 63

DATA ACCEPTED BY AAVSO: 812

CLEAR NIGHTS THIS MONTH: 22

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YOUR DATA WAS USED IN:



ATel #16234

"Early photometry of Nova Her 2025  
confirms classical nova classification"

Your telescope: 4 data points



AAVSO Database

Nova Herculis 2025 light curve

Your telescope: 5 nights of data

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ACHIEVEMENTS



First Light



Century Club (100 observations)



Nova Hunter (observed a nova)



Night Owl (500 observations)



Supernova Hunter (observe 3 SNe)

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## ## Accessibility Features Checklist

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### VISUAL ACCESSIBILITY:

- ✓ Full VoiceOver (iOS) support
- ✓ Full TalkBack (Android) support
- ✓ Dynamic text sizing (follows system)
- ✓ High contrast mode
- ✓ Reduced motion mode
- ✓ Color-blind safe palette  
(no red-green only information)
- ✓ All images have alt text
- ✓ All charts have text/table alternatives
- ✓ Dark mode (reduces glare)
- ✓ OpenDyslexic font option

### MOTOR ACCESSIBILITY:

- ✓ Single-tap navigation (no multi-finger required)
- ✓ Switch control fully supported
- ✓ Voice control fully supported  
(unique labels on all interactive elements)
- ✓ Eye gaze compatible (large targets)
- ✓ No time-limited interactions
- ✓ No required motion gestures
- ✓ Minimum 44×44pt touch targets

### COGNITIVE ACCESSIBILITY:

- ✓ Consistent navigation structure
- ✓ Plain language throughout
- ✓ Reading level selector  
(child, standard, expert)
- ✓ Jargon definitions on tap
- ✓ Progress auto-saved
- ✓ No unexpected interruptions
- ✓ Reduce visual complexity mode
- ✓ No auto-playing content

### HEARING ACCESSIBILITY:

- ✓ No audio-only information

- ✓ All sounds have visual equivalents
- ✓ Captions on all video content
- ✓ Vibration alternatives for audio alerts

#### NOVEL ACCESSIBILITY FEATURES:

- ✓ Audio descriptions of all data  
(three depth levels)
- ✓ Sonification of light curves
- ✓ Haptic light curves
- ✓ Spatial audio sky map  
(hold phone up, hear what's there)
- ✓ Patience mode (no rushing, ever)
- ✓ Proxy observation support  
(caregiver setup, member receives results)

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## # The Network

### ## Node Requirements

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#### MEMBER PROVIDES:

- Seestar S50 or S30 (or future supported scopes)
- Computer (Windows, Mac, or Linux)  
that can be left on overnight
- Home WiFi network
- Power outlet

#### BOUNDLESS SKIES PROVIDES:

- Node software (free download)
- All scheduling and coordination
- All data processing
- All scientific submissions
- Full dashboard access
- Credit on all observations

#### MINIMUM TECHNICAL REQUIREMENT:

- Ability to follow a setup guide
- That's it

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### ## Node Tiers

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#### TIER 1: STANDARD NODE

Equipment: Seestar (any model)  
Filter: Broadband (CV/CR designation)  
Limiting mag: ~14  
Precision: ~0.05-0.1 mag  
Value: Coverage, redundancy, bright targets

This is the primary network node.  
Anyone with a Seestar can contribute.

#### TIER 2: FILTERED NODE (future)

Equipment: 6-8" scope + mono camera + BVRI filters  
Filter: Johnson-Cousins BVRI  
Limiting mag: ~17-18  
Precision: ~0.01-0.02 mag  
Value: Precise multi-band photometry,  
fainter targets, color evolution

Added when network matures and  
members want to upgrade contribution.

#### TIER 3: SPECTROSCOPY NODE (future)

Equipment: Tier 2 + spectrograph  
Added for brightest targets (mag < 10)  
Enables classification confirmation  
Rare, high-value nodes

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### ## Geographic Strategy

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#### PRIORITY LONGITUDES (for 24-hour coverage):

Zone 1: Americas (UTC-8 to UTC-4)  
Target: 30% of nodes  
Coverage: 01:00-09:00 UTC

Zone 2: Europe/Africa (UTC-0 to UTC+3)  
Target: 30% of nodes  
Coverage: 20:00-04:00 UTC

Zone 3: Asia/Pacific (UTC+8 to UTC+12)



Target: 30% of nodes  
Coverage: 11:00-19:00 UTC

Zone 4: Middle East/India (UTC+4 to UTC+7)  
Target: 10% of nodes  
Coverage: Gap-filling

With this distribution:

Any target can be observed continuously  
for 24 hours by different nodes.

No target is ever "out of reach."

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# The Organization

## Funding Model

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PRIMARY FUNDING SOURCES:

Grants (target 70% of budget):

- NSF (National Science Foundation)  
Education and public outreach programs
- NASA citizen science program grants
- Simons Foundation (astronomy + education)
- Chan Zuckerberg Initiative (science)
- Google.org
- Apple accessibility grants
- Foundation for the Blind
- Christopher & Dana Reeve Foundation
- Astronomical Society of the Pacific

In-Kind Donations (target 20% of value):

- AWS/Azure/Google Cloud nonprofit credits  
(potentially \$5,000-25,000/year free compute)
- ZWO telescope donations for loan program
- Equipment from retiring amateur astronomers
- Software licenses

Individual Donations (target 10% of budget):

- Monthly donors
- One-time donations

- Memorial/tribute donations  
("A telescope observed in memory of...")
- Node naming rights for major donors

#### NOT RELYING ON:

- Member subscription fees  
(members contribute time, not money)
- Hardware sales  
(not a hardware company)
- Advertising  
(never)

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#### ## Budget Projections

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##### PHASE 0 (months 1-3): ~\$100 total

Cloud server: \$20/month  
Domain: \$12/year  
Testing equipment: \$0 (borrowed)

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Total: ~\$100

##### PHASE 1 (months 3-9): ~\$500-1,000 total

Cloud server: \$30-50/month  
Image storage: \$10-20/month  
Domain + email: \$15/month  
Development tools: \$0 (free tiers)  
Nonprofit filing: \$100-300 one-time

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Total: ~\$500-1,000

##### PHASE 2 (months 9-18): Grant-funded

First grant target: \$5,000-25,000

##### Covers:

- Cloud infrastructure
- App development tools
- First telescope loans
- Travel to conferences
- Legal/accounting

#### SUSTAINABILITY TARGET:

\$50,000/year in grants by year 2  
\$200,000/year by year 3

Self-sustaining on grants + donations by year 3  
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## # Partnerships

### ## Priority Partnerships

#### ### ZWO (Hardware Partner)

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##### ASK:

- Official API documentation for local protocol
- Telescope donations for loan program
- Co-marketing ("Seestar powers Boundless Skies")
- Feature inclusion in Seestar app

##### OFFER:

- Marketing story ("Seestar does real science")
- Data showing Seestar scientific capability
- Community of engaged Seestar users
- Credibility via AAVSO partnership

STATUS: Email to be sent this week

...

#### ### AAVSO (Science Partner)

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##### ASK:

- Multi-node observer code setup
- Programmatic submission API access
- Data quality guidance
- Possible fiscal sponsorship  
(while nonprofit status pending)
- Endorsement

##### OFFER:

- Significant increase in AAVSO submissions
- New demographics of observers
- Automated high-quality pipeline
- Accessible astronomy alignment with  
their own outreach mission

STATUS: Email to be sent this week

...

### ### Disability Organizations

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#### TARGET PARTNERS:

- American Foundation for the Blind
- National Federation of the Blind
- Christopher & Dana Reeve Foundation
- United Spinal Association

#### ASK:

- User testing pool
- Community access for recruitment
- Co-grant applications
- Endorsement

#### OFFER:

- Genuine astronomy access for their communities
- Novel accessible technology
- Named participation in real science
- Something genuinely new

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### ### Educational Institutions

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#### TARGET PARTNERS:

- Perkins School for the Blind
- Schools for deaf students
- Hospital-based education programs
- University astronomy departments

#### ASK:

- Node hosting (place telescope at school)
- Student participants
- Academic credibility

#### OFFER:

- Free network access
- Curriculum materials
- Student names on scientific outputs
- Real science participation

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## # Technology Stack

### ## Cloud Infrastructure

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HOSTING: AWS or DigitalOcean  
(AWS preferred when nonprofit credits obtained)

#### COMPUTE:

- Application server: Python/FastAPI
- Task queue: Celery + Redis  
(for async photometry processing)
- Database: PostgreSQL
- Cache: Redis

#### STORAGE:

- Database: PostgreSQL on managed service
- Image storage: S3-compatible  
(Cloudflare R2 preferred — no egress fees)
- Backup: Daily automated

#### ALERT PROCESSING:

- Alert ingestion: Python scripts  
polling ALerCE, ATLAS, ASAS-SN APIs
- Processing: Celery async tasks
- Storage: PostgreSQL targets table

#### SCHEDULING:

- Nightly scheduling job (Celery beat)
- Real-time interrupt handling
- Node dispatch via REST API

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### ## Node Software

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LANGUAGE: Python 3.10+  
FRAMEWORK: None (simple scripts + services)  
KEY DEPENDENCIES:  
→ seestar\_alp (telescope control)  
→ astropy (FITS, coordinates, time)

- photutils (aperture photometry)
- astroquery (catalog queries)
- requests (HTTP client for cloud API)
- watchdog (file system monitoring)

#### PACKAGING:

- PyInstaller (creates single executable)
- NSIS (Windows installer)
- DMG creator (Mac installer)
- AppImage (Linux)

#### SERVICE MANAGEMENT:

- Windows: Windows Service via pywin32
- Mac: launchd plist
- Linux: systemd service

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#### ## Mobile App

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FRAMEWORK: Flutter (cross-platform)

PLATFORMS: iOS, Android, Web (progressive web app)

LANGUAGE: Dart

#### KEY PACKAGES:

- fl\_chart (light curve visualization)
- flutter\_tts (text to speech for audio descriptions)
- vibration (haptic patterns)
- sensors\_plus (accelerometer for sky map)
- firebase\_messaging (push notifications)
- shared\_preferences (local settings)
- http (cloud API client)

#### ACCESSIBILITY:

- Semantics widgets throughout
- ExcludeSemantics where needed
- MergeSemantics for grouped content
- Tooltip on all interactive elements
- Focus management for keyboard/switch nav
- Tested with VoiceOver and TalkBack

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#### ## Data Pipeline

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PLATE SOLVING: ASTAP (local, fast, free)  
PHOTOMETRY: Photutils (Python, well-maintained)  
CATALOG ACCESS: Astroquery  
→ AAVSO comparison sequences  
→ Gaia DR3  
→ ATLAS Refcat2  
TIME CONVERSION: Astropy (BJD\_TDB)  
FITS HANDLING: Astropy.io.fits  
QUALITY CONTROL: Custom Python  
AAVSO SUBMISSION: AAVSO WebObs API  
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## # Development Roadmap

### ## Phase 0: Proof of Concept (Months 1-3)

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GOAL: Prove the science works

#### MILESTONES:

M1: seestar\_alp controlling a Seestar  
via Alpaca API (Week 2)

M2: Images retrieved from Seestar  
via SMB share automatically (Week 3)

M3: Photometry pipeline producing  
calibrated magnitudes from  
Seestar images (Week 4-6)

M4: First AAVSO-accepted observation  
from automated pipeline (Month 2-3)

#### DELIVERABLES:

- Working single-node pipeline
- First AAVSO submission
- Validated photometric precision
- Proof the science works

#### SUCCESS CRITERIA:

One AAVSO-accepted automated observation

from a Seestar with magnitude agreeing within 0.15 mag of known value.

BUDGET: ~\$100

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## ## Phase 1: Core System (Months 3-9)

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GOAL: Build the real system

### MILESTONES:

M5: Cloud server with alert ingestion, scoring, scheduling (Month 4)

M6: Node software as installable service (Month 5)

M7: End-to-end: alert → schedule → observe → process → submit (Month 6)

M8: First Flutter app (Month 7)  
Basic dashboard, accessibility-first

M9: 3-5 beta nodes running (Month 8)

M10: First multi-node coordinated observation (Month 9)

### DELIVERABLES:

- Working multi-node system
- Basic accessible mobile app
- Regular AAVSO submissions
- 3-5 beta members

BUDGET: ~\$500-1,000

(covered by initial donations/personal)

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## ## Phase 2: Launch (Months 9-18)

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GOAL: Public launch, grow to 50 nodes



#### MILESTONES:

M11: Nonprofit status obtained (Month 10)

M12: First grant received (Month 11-12)

M13: App on iOS and Android App Stores  
(Month 12)

M14: 25 active nodes (Month 14)

M15: First ATel with Boundless Skies data  
(Month 15)

M16: 50 active nodes (Month 18)

M17: First academic paper mentioning  
Boundless Skies (Month 18)

#### DELIVERABLES:

- Public-facing network
- Accessible app in App Stores
- Formal nonprofit status
- Grant funding operational
- Regular scientific outputs
- ZWO partnership established
- AAVSO formal partnership

BUDGET: First grant (\$5,000-25,000)

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## Phase 3: Growth (Months 18-36)

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GOAL: 200 nodes, recognized instrument

#### MILESTONES:

M18: 100 active nodes (Month 24)

M19: Network paper submitted (Month 24)

M20: 200 active nodes (Month 30)

M21: First disabled astronomer's data  
cited in peer-reviewed paper (Month 30)

M22: LIGO/gravitational wave follow-up  
participation (Month 36)

M23: Telescope loan program active  
(10+ scopes loaned) (Month 36)

DELIVERABLES:

- Recognized scientific instrument
- Published network paper
- Active telescope loan program
- Multiple grant streams
- Staff hired (part-time)
- Full accessibility feature set
- Multiple smart telescope types supported

BUDGET: \$50,000-200,000/year in grants

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# Risk Assessment

## Technical Risks

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RISK: seestar\_alp breaks when ZWO updates firmware

PROBABILITY: Medium (happens periodically)

IMPACT: Temporary node outages

MITIGATION:

- Monitor ZWO firmware releases
- Test firmware updates before pushing to network
- Engage with seestar\_alp developer community
- Contribute to seestar\_alp maintenance
- Version-lock firmware when stable

RISK: Seestar photometry precision insufficient

PROBABILITY: Low (preliminary evidence positive)

IMPACT: AAVSO rejects data, science value low

MITIGATION:

- Test BEFORE building anything else
- This is Phase 0 milestone 1
- If insufficient: pivot to detection  
confirmation role rather than precision  
photometry, target brighter stars only

RISK: Member computers going to sleep

PROBABILITY: High (if not configured)

IMPACT: Node offline during night

MITIGATION:

- Installer configures sleep settings automatically
- Test: keep-awake script during observing hours
- Documentation: clear instructions for manual check
- Monitoring: detect when nodes go dark unexpectedly

RISK: SMB share access issues

PROBABILITY: Low-Medium (router dependent)

IMPACT: Images not retrieved

MITIGATION:

- Test SMB access during setup
- Alternative: direct HTTP download if SMB fails
- Troubleshooting guide for common router issues

...

## ## Organizational Risks

...

RISK: ZWO changes Seestar protocol making  
seestar\_alp incompatible

PROBABILITY: Low-Medium

IMPACT: High (entire network relies on this)

MITIGATION:

- Pursue official ZWO API partnership
- Contribute to seestar\_alp
- Build protocol abstraction layer  
(easy to swap in new driver)
- Diversify to other smart telescopes  
(DWARF, Unistellar) over time

RISK: Not enough nodes in key longitudes

PROBABILITY: Medium

IMPACT: Coverage gaps in continuous monitoring

MITIGATION:

- Active recruitment in underrepresented timezones
- Partner with international astronomy clubs
- Consider telescope loan program for key sites
- Design scheduler to handle coverage gaps gracefully

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## # Success Metrics

## ## Phase 0 (Month 3)

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- ✓ One AAVSO-accepted automated observation
- ✓ Photometric precision validated
- ✓ End-to-end pipeline working
- ✓ ZWO email sent and response received
- ✓ AAVSO email sent and response received

...

## ## Phase 1 (Month 9)

...

- ✓ 5 active nodes in 3+ countries
- ✓ 100+ AAVSO submissions from network
- ✓ First accessible app version live
- ✓ Nonprofit filing submitted
- ✓ First grant application submitted
- ✓ Network producing data every clear night

...

## ## Phase 2 (Month 18)

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- ✓ 50 active nodes
- ✓ 10+ countries represented
- ✓ 1,000+ AAVSO submissions
- ✓ App on iOS and Android
- ✓ First ATel with network data
- ✓ Nonprofit status received
- ✓ First grant received
- ✓ First disabled astronomer testimony documented
- ✓ ZWO partnership established

...

## ## Phase 3 (Month 36)

...

- ✓ 200 active nodes
- ✓ 25+ countries
- ✓ 10,000+ AAVSO submissions
- ✓ Network paper published
- ✓ Disabled astronomer's data cited

- ✓ 10+ telescopes in loan program
- ✓ \$50,000+ annual grant funding
- ✓ Part-time staff
- ✓ Recognized as scientific instrument

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## # Final Statement

Boundless Skies is not a simulation of astronomy. It is not a game about astronomy. It is not a visualization of someone else's data.

It is astronomy.

Real telescopes. Real targets. Real measurements. Real publications. Real science.

The only thing that changes is who gets to do it.

The answer, with Boundless Skies, is everyone.

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\*Boundless Skies | [boundlesssskies.org](http://boundlesssskies.org)\*

\*Founded 2025 by Eli Goldfine and Scott Mellis\*

\*"The night sky belongs to everyone."\*